Q22 Deeper Project Suggestions

1. **Harness ablation study:** take one fixed LLM (via API or local weights) and build three harnesses around it — single call, tool-augmented single pass, full ReAct-style loop — on the same small task suite (e.g., 20 file-editing tasks). Report success rate, cost, and trajectory length for each. Quantify how much of the "agent capability" is the loop.
2. **Interface design experiment:** in the spirit of SWE-agent, give your loop two different tool surfaces for the same environment (raw shell vs. a curated edit/search API with guardrails) and measure the gap. Vary feedback verbosity and error-message quality; identify which harness detail moves the success rate most per engineering hour.
3. **Context rot instrumentation:** instrument an agent loop to log context length, position of the critical instruction, and per-step accuracy over a 50-step task. Reproduce a position-dependent degradation curve in your own loop and test two mitigations (compaction vs. instruction re-injection at the context head) against it.
4. **Stopping-criterion zoo:** implement one agentic harness with swappable stopping rules (step cap, token budget, model-declared finish, external verifier). Run each on tasks with known ground truth and plot success rate vs. mean steps and cost. Characterize when a model-declared "finish" is trustworthy and when an external verifier is mandatory — a concrete entry point into Q24's evaluation contracts.